

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ 4 SET B

SUMMER SEMESTER, 2024-2025

DURATION: 25 MINUTES

FULL MARKS: 20

CSE 4205: Digital Logic Design

Programmable calculators are not allowed.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

ID:

1. Draw a three-bit bidirectional shift register. Clearly show your choice of flip-flops and label the circuit properly. The register must shift **right** when $M = 1$. Trace the data path for this mode.

5
(CO3)
(PO2)

2. A Type-A flip-flop has two inputs and can exhibit the race-around condition for one input combination. A Type-B flip-flop has one input and is commonly used in counters and for controlled toggling. You must implement the behavior of Type B using Type A. 10
(CO3)
(PO2)

Complete the conversion using the following steps.

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|--|-----|--|-----|
| (a) Identify Types A and B. | [1] | (d) Construct the conversion table. | [2] |
| (b) Write the Type-B characteristic table. | [2] | (e) Derive the Type-A input equations. | [2] |
| (c) Write the Type-A excitation table. | [2] | (f) Draw the implementation. | [1] |

3. Design a three-bit synchronous sequential circuit using T flip-flops that cycles, in ascending order, through all three-bit values satisfying $x \bmod 2 = 1$, and then returns to the first value. Let the state be $Q_2Q_1Q_0$.

5
(CO5)
(PO3)

Complete the state/excitation table and derive minimized equations for T_2, T_1 , and T_0 . You do **not** need to draw the final logic circuit.

For reference, the T-flip-flop excitation table is

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

[Bonus: 2] After implementing your equations and connecting all flip-flops to the same clock, the circuit is powered up in state 000 and does not enter the required sequence. Explain why, and state one practical correction.